

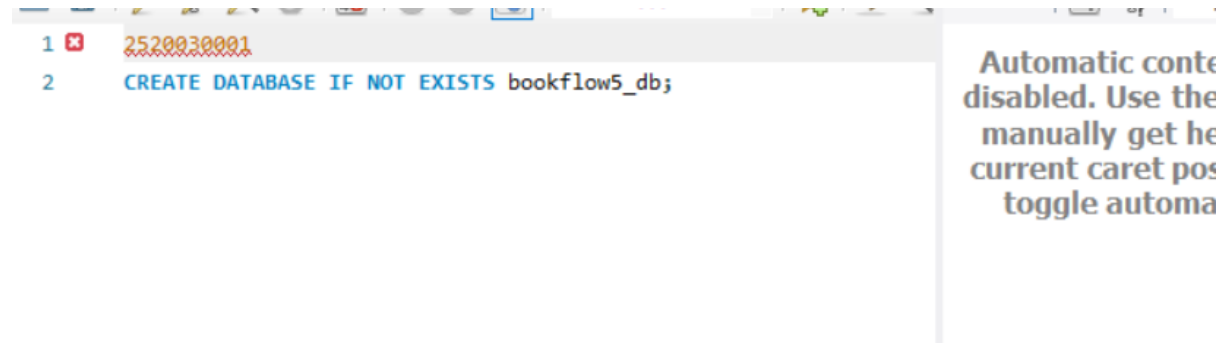
Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030001

NAME: MYDAM ASWITHA PATEL

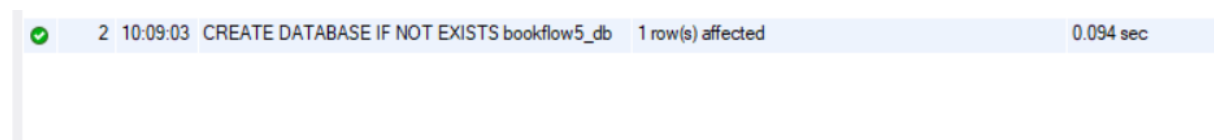
Query 1: Create the database



The screenshot shows a SQL query editor with two lines of code. Line 1 is a redacted username. Line 2 is the SQL command to create a database. To the right of the editor, there is a tooltip that reads: "Automatic content disabled. Use the manually get the current caret position to toggle automatic content." The code in the editor is as follows:

```
1 2520030001
2 CREATE DATABASE IF NOT EXISTS bookflow5_db;
```

Output: -



The screenshot shows the execution results of the first query. It is a single row with the following data:

✓	2	10:09:03	CREATE DATABASE IF NOT EXISTS bookflow5_db	1 row(s) affected	0.094 sec
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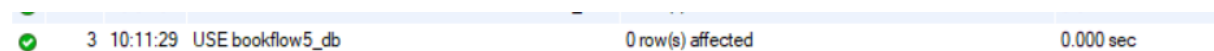
Query 2: Select the database



The screenshot shows a SQL query editor with two lines of code. Line 1 is a redacted username. Line 2 is the SQL command to use a specific database. To the right of the editor, there is a tooltip that reads: "Automatic content disabled. Use the manually get the current caret position to toggle automatic content." The code in the editor is as follows:

```
1 2520030001
2 USE bookflow5_db;
```

Output: -



The screenshot shows the execution results of the second query. It is a single row with the following data:

✓	3	10:11:29	USE bookflow5_db	0 row(s) affected	0.000 sec
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Query 3: Create the books table

2520030001-Aswitha

```
CREATE TABLE books (  
  book_id INT AUTO_INCREMENT PRIMARY KEY,  
  title VARCHAR(255) NOT NULL,  
  isbn VARCHAR(13) NOT NULL UNIQUE,  
  published_year INT,  
  CONSTRAINT chk_published_year CHECK (published_year < 2027)  
);
```

Output: -

✓	4	10:18:46	CREATE TABLE books (book_id INT AUTO_INCR...	0 row(s) affected	0.125 sec
---	---	----------	---	-------------------	-----------

Query 4: Create the members table

```
);
```

2520030001-Aswitha

```
CREATE TABLE members (  
  member_id INT AUTO_INCREMENT PRIMARY KEY,  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(150) NOT NULL UNIQUE  
);
```

[Context Help](#) [Snippets](#)

Output: -

✓	5	10:23:27	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec
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Query 5 (verification): Inspect the table structures

```
);
```

```
DESCRIBE books;
```

Output: -

✓	6	10:26:25	DESCRIBE books	4 row(s) returned	0.047 sec / 0.000 sec
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Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		
	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(255)	NO		NULL	
	isbn	varchar(13)	NO	UNI	NULL	
	published_year	int	YES		NULL	

Query 6: Insert 3 books

```
2520030001-Aswitha
INSERT INTO books (title,isbn,published_year) VALUES
('The Alchemist','9780061122415',1988),
('The autobiography','9780132350884',2005),
('Harry Potter','9780735211292',1997);
```

Output: -

✓	8	10:33:11	INSERT INTO books (title,isbn,published_year) VAL...	3 row(s) affected	Records: 3	Duplicates: 0	Warning...	0.031 sec
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Query 7: Insert 3 members

```
24      ( 'Harry Potter' , '9780735211292' ,1997 );
25 •    INSERT INTO members (full_name,email) VALUES
26      ('Aswitha Patel','aswitha@gmail.com'),
27      ('Adhritha','adhritha@gmail.com'),
28      ('Nihitha','nihitha@gmail.com');
```

Output: -

✓	9	10:38:04	INSERT INTO members (full_name,email) VALUES (...	3 row(s) affected	Records: 3	Duplicates: 0	Warning...	0.047 sec
---	---	----------	---	-------------------	------------	---------------	------------	-----------

Query 8: Display

```
SELECT * FROM books;
```

Output: -

10 10:40:26 SELECT * FROM books LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	The autobiography	9780132350884	2005
3	Harry Potter	9780735211292	1997
NULL	NULL	NULL	NULL

```
SELECT * FROM members;
```

11 10:41:15 SELECT * FROM members LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec

member_id	full_name	email
1	Aswitha Patel	aswitha@gmail.com
2	Adhritha	adhrittha@gmail.com
3	Nihitha	nihitha@gmail.com
NULL	NULL	NULL

Query 9: Constraint Tests (These MUST Fail)

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Output: -

12	10:47:42	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 1062. Duplicate entry '9780061122415' for key...	0.047 sec
----	----------	---	--	-----------

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Output: -

14	10:50:07	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null	0.000 sec
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Test 3: Future publication year (CHECK violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '8888888888888', 2030);
```

Output: -

15	10:52:21	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.	0.000 sec
----	----------	---	--	-----------

Test 4: Duplicate email (UNIQUE violation)

```
INSERT INTO members (full_name, email)
VALUES ('Adhritha Reddy', 'adhritha@gmail.com');
```

Output: -

19	10:57:15	INSERT INTO members (full_name, email) VALUES ('Adhritha Reddy', 'adhritha@gmail.com')	Error Code: 1062. Duplicate entry 'adhritha@gmail.com' for key 'members_email'	0.031 sec
----	----------	--	--	-----------